EN.601.762 Project Proposal

Verified Doom Level Generation: A Playability-Checked Intermediate Representation for Language Agents

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Abstract. Language models can describe a good Doom level, but the maps they produce often cannot be finished: rooms overlap, props block doorways, lifts strand the player, and keys sit behind their own locks. Our pilot maps scored well on path-clarity heuristics yet failed fair play in GZDoom because of exactly these geometry errors. We argue the bottleneck is not the prompt but the absence of a *checkable contract* between the model and the engine. We contribute (1) an intermediate representation (IR) that encodes a level as a *critical-path graph* (start, teach, key, lock, peak, exit, with precedence and optional branches) plus typed geometry (true L-shaped rooms, doors, lifts, cover props, monster closets); (2) a deterministic compiler from IR to UDMF that is *verified*: before writing a PWAD it runs a hull and walkability gate (room and closet overlap, hollow or portal-blocking props, incomplete door and lift openings, and a breadth-first search at Doom's 16-unit player radius from spawn to exit) and refuses to emit failing maps; (3) a layered metric suite that separates *playable* (binary, from the gate), *trajectory* (key-before-lock, weapon-before-peak, early-combat cap, hub return, landmark clearance), and *design* (a five-pillar rubric), so that a pretty but broken level scores zero; and (4) a verifier-guided repair loop in which gate and trajectory failures are returned to the generator as structured, localized error messages instead of free-form critique. The generator itself is interchangeable: we evaluate local Ollama models and a rule-based baseline under the same IR, compiler, and metrics, so gains can be attributed to the representation and verifier rather than to prompting a particular model. Inference runs locally (RTX 5070, 16 GB) with free assets (Freedoom).

Grounding. The IR follows Riedl's view that in games, story is progression through space: experience management and GAME FORGE encode intended play as a plot graph of ordered events with optional branches, realized as "islands" (required rooms) joined by "bridges" (corridors), and we enforce that optional wings never sit on the only route between plot nodes. The repair loop follows Rocktäschel's framing of open-ended systems as producing novel but learnable artifacts through cheap, strict environments and self-improvement loops: our gate is the learnability filter, structured failures are the improvement signal, and a stretch goal is a quality–diversity archive of level patterns that keeps only maps passing the gate.

Research questions.

- **RQ1:** Does the critical-path IR raise the rate of finishable maps over generating raw room/connection JSON with the same model?
- **RQ2:** Does structured verifier feedback repair more failures per iteration than free-form LLM or VLM critique?
- **RQ3:** Do the automatic playable and trajectory metrics agree with human fair-play outcomes better than design heuristics alone?

Evaluation. Primary metric: *verified-finishable rate* (passes the gate and the trajectory checks), confirmed on a subset by human fair play in GZDoom without cheats. Secondary: repairs needed to reach a pass, trajectory and design scores, and diversity across briefs. Baselines: rule-based generator; LLM emitting flat JSON without the IR; IR without the gate; IR with gate but free-form critique. Ablations remove one component at a time (IR, gate, structured feedback, classic-map few-shot references).

Planned mid deliverables (Oct 19)

- IR schema and validator; UDMF compiler with L-rooms, doors, keyed locks, lifts, props, and closets.
- Hull and walkability gate integrated as a hard build step, with unit tests on known failure cases from our playtests.

- Layered metrics (playable / trajectory / design) on at least 10 generated maps, with a failure gallery that labels which check caught each bug.
- End-to-end demo: brief $\rightarrow$ IR $\rightarrow$ verified `.wad` in UDB and GZDoom.
- Mid report (NeurIPS 2026 format, $\leq$4 pages) and in-class presentation.

Planned final deliverables (Dec 9 poster; Dec 15 report)

- Key-aware walkability (locks closed until the key is reached) and two-way lift checks in the gate.
- Verifier-guided repair loop and comparison against free-form critique (RQ2).
- Baselines and ablations on a fixed set of 30+ briefs (RQ1).
- Small human fair-play study measuring metric–outcome agreement (RQ3).
- Released IR, compiler, gate, metric code, and the brief set; final poster and NeurIPS-style report ($\leq$8 pages).